

Engaging coastal communities with climate change

Organisation: Defra / JBA Consulting

Case study focus

Raising Awareness / Community engagement / Coastal flooding / Coastal erosion

This case study summarises a research project that identified effective approaches to engaging coastal communities and the third sector with UKCP09 projections.

Recognising the crucial role of community participation in planning for and managing adaptation to coastal change, Defra explored how to better understand and engage people who are living in areas of high risk. The project developed approaches to communicate the UKCP09 projections to people living in areas where the effects of sea level rise, flooding and coastal erosion will be most felt.

What was it used for?

The project is an example of **stakeholder engagement** with individuals and organisations that may be not generally well informed on wider environmental matters, or on the impacts of climate change on their communities. The project also helped to identify how key messaging from UKCP09 is best used for the purposes of **awareness raising** at a local level and in light of the audience it is being targeted at.

Which UKCP09 products were used?

[Marine & coastal projections report](#)

[Climate change projections report](#)

[Maps & key findings](#)

Approach and how UKCP09 was used

In order to investigate the most effective approach to engagement of coastal communities, the research was divided into three phases:

- Phase 1: This was the baseline research and planning for the overall project including desk research to establish existing approaches to climate change communication and engagement.

- Phase 2: The second phase of the project was the delivery of 12 focus groups at locations considered to be at long term risk from climate change, discussing and gaining insights into current levels of knowledge and attitudes towards climate change and the UKCP09 projections.
- Phase 3: The third phase involved evaluating feedback from the focus groups and all other project research to provide the basis for guidance and recommendations on effective methods of engagement with citizens and groups in at-risk communities.

Using UKCP09: Phase 1

UKCP09 was used extensively in Phases 1 and 2 of the project.

In Phase 1, the projections were used to define the key messages that would be presented to the focus groups. In order to do this, the project team researched the [Marine & coastal projections report](#), [Climate change projections report](#) and [Maps & key findings](#) pages. They looked for key projections that would affect coastal settlements, such as erosion and flooding.

The study identified several key messages that the project team could communicate to coastal communities, these included:

- **Sea level rise** - the UKCP09 projections for global sea level rise give an estimated range for increases of between 18 and 59 cm by the end of the century (2090-2099). The central estimates for sea level change at four locations around the UK are supplied for the three emissions scenarios, Low, Medium and High. These give a range of central estimates across all scenarios of a rise of between 19.6 and 43.3 cm from 1990 to 2080. The full range of estimates over all emissions scenarios of absolute global sea level rise are given as 11.6 to 75.8 cm for the period 1990 to 2090-2099.
- **Wave height** - the UKCP09 marine projections include changes in the winter mean wave height that are projected to be between -35 and +5 cm, with changes in the annual maxima projected to be between -1.5 and +1 m.
- **Water temperature** - the most noticeable and robust impact of climate change on shelf seas around the UK in the next century is temperature increase. This amounts to about 1.5°C – 4°C, depending on location, by the end of the 21st century (for the medium emission scenario) with the increase being larger in shelf seas than the open ocean.
- **Wetter/Drier seasons** – the study also researched the '[Climate change projections report](#)' in order to investigate potential temperature and precipitation changes. The report indicates a warming across the UK of between 2° and 3°C by 2050 (relative to 1961-1990), an increase in winter rainfall of 10-20% (lower in the uplands), and a decrease in summer rainfall of 10-20% (more in the south west).

The project team found that the UKCP09 scenarios do not provide outputs for possible changes in year-to-year climatic variability, such as altered frequencies of successive dry winters.

Another variable that the team considered investigating was **storminess**. While the effects of climate change on storminess are not well understood, it was still possible that the strength and frequency of coastal storms are being affected by climate change. The UKCP09 scenarios do not provide outputs for changes in wind speeds. It is however possible to investigate this using the [11-member RCM](#).

After identifying the key messages, focus group presentations were prepared in order to investigate the most effective way to communicate these messages.

Using UKCP09: Phase 2

Phase 2 of the project saw the key messages from UKCP09 communicated to focus groups in areas that are at risk from some of the potential coastal impacts of climate change. The project team presented the information in three different presentations, to 12 focus groups in coastal communities around England. Each presentation was designed differently and the reactions of the groups were recorded.

- **Presentation 1:** The first presentation tested was one which presented the projections directly from UKCP09. It was written using factual language, supported with image boards using terms and images derived from the UKCP09 website.
- **Presentation 2:** The second presentation tried to convey similar messages but was written in a much more engaging and simpler way. It used graphics that were associated with climate change projections taken from the Met Office website.
- **Presentation 3:** Finally a third way of presenting the messages was considered. This presentation was based on simple 'message board' images and accompanying 'headline messages'.

Examples of the three styles of presentations can be seen in Figure 1.

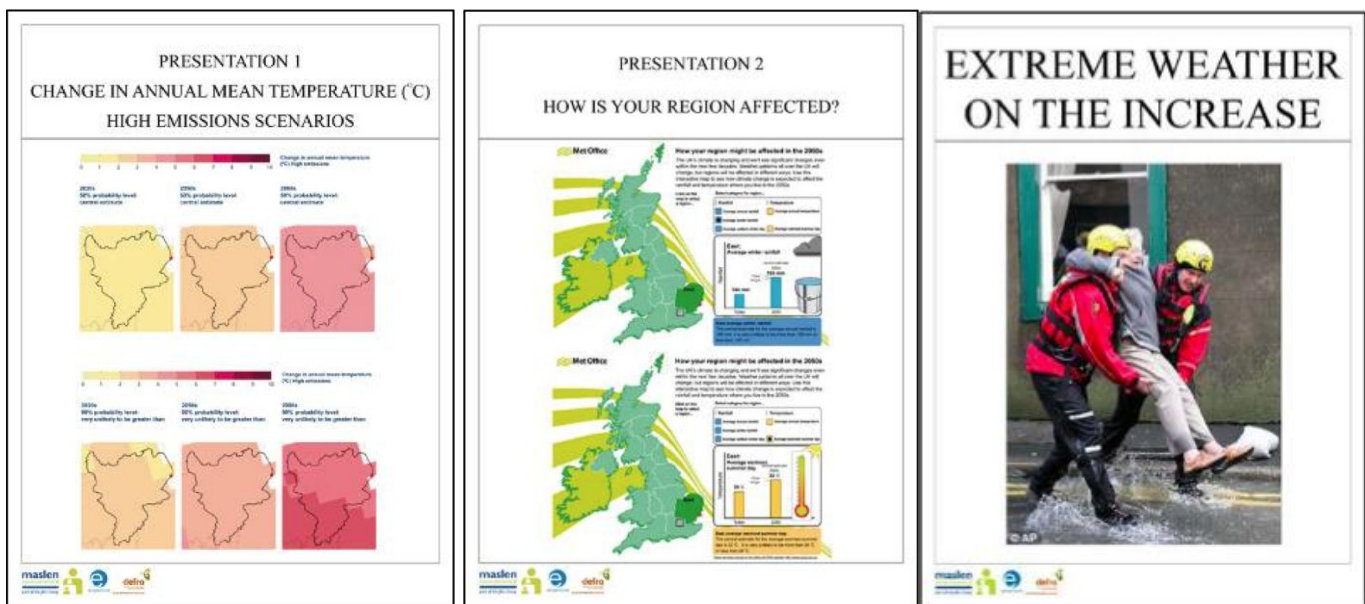


Figure 1: The three contrasting presentation styles used in the project. From left to right: Presentation 1 engaged directly with the projections and presented images taken directly from the UKCP09 site; Presentation 2 used similar key messages but displayed them using simpler language and more basic images; Presentation 3 used advertising-style 'message boards'.

Results and interpretation

Phase 3 of the project involved analysing the results of the feedback on the presentations from the focus groups.

The presentations were tested against the overall goal of encouraging people to understand climate change and to take action to mitigate and adapt to it. The study found that participants saw Presentation 1 as being the hardest to understand and least likely to encourage action. Presentation 2 was found to be more comprehensible, but that participants were still quick to question the validity of the results. The third presentation included message boards that connected with the focus groups on an emotional level; including those that had pictures of animals. These resonated with participants more than the other presentations, but included less detail about the specific impacts of climate change.

Because there is still a considerable amount of scepticism about climate change projections amongst the general public, it was hard to encourage them to understand that UKCP09 projections were more than just 'opinion'. They found that one of the challenges for communicating UKCP09 projections was establishing their credibility and explaining that 'uncertainty' does not mean that the projections are incorrect or conjecture, but is inevitable in science.

The study also found that communications efforts should focus on overcoming some of the key barriers to acceptance of climate change. In this way, it is thought people will be more likely to take action to mitigate and adapt. Figure 2 shows the connection between awareness and understanding and the pathways to action.

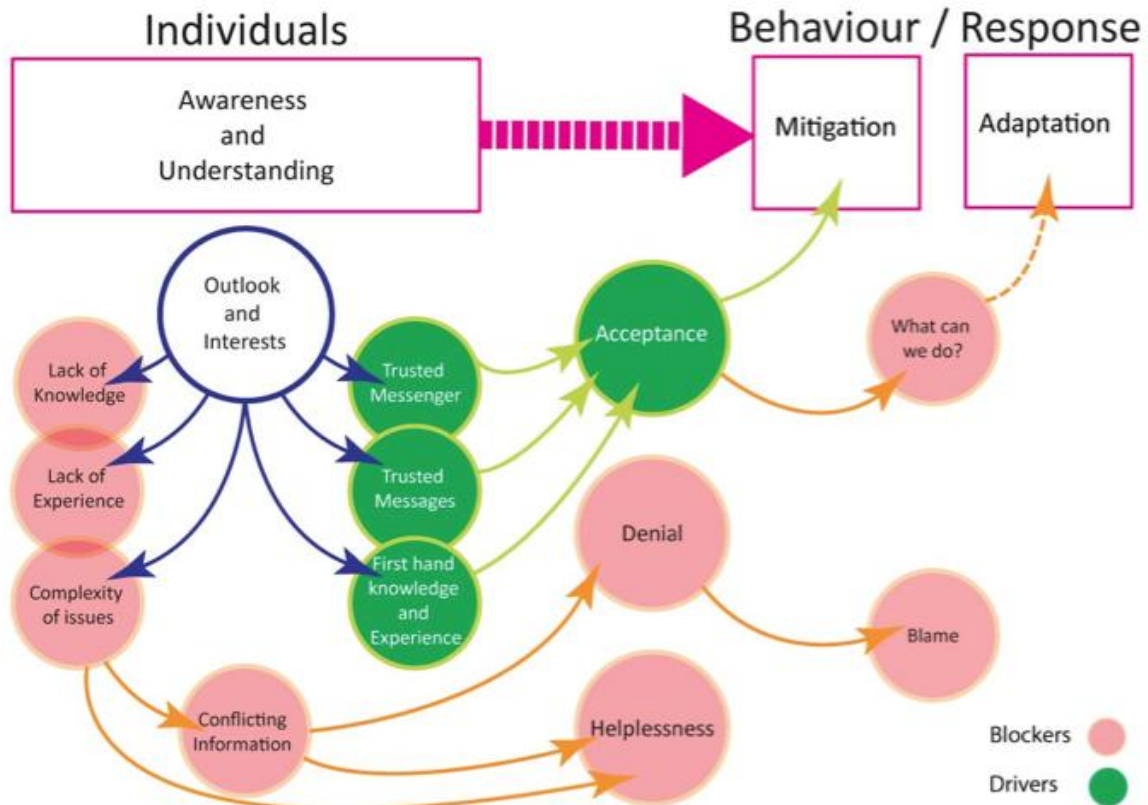


Figure 2: Connection between acceptance of climate change and action

Communicate to your audience

The study found that the UKCP09 projections were useful to identify the key messages about climate change, but that it was vital to communicate them in a way that would resonate with the target audience.

In light of this, the study recommended that:

1. Any messages to the public should be written in plain English and not be too technical or contain too many figures. They should be informative and factual. Many people are sceptical about the reliability of evidence following negative climate change news coverage.
2. Where possible, support the use of probabilities and projections with observed facts and examples. While care has been taken to ensure the information provided in UKCP09 is technically correct, it may be helpful to translate or tailor some of the more scientific content for non-specialist audiences.
3. The link between climate change, sea level rise and flooding can be explained, but may not be fully accepted by many people. Some refuse to believe in global warming.

4. By focusing on communicating the impacts of climate change such as flooding and coastal erosion rather than the causes, this will assist in avoiding unnecessary public confusion and scepticism about the causes of climate change. Examples from coastal erosion and flooding incidents should be drawn.
5. People prefer simple pictures that can be easily digested to sophisticated imagery that involves more intellectual processing. Images of animals, for example, resonate with people and could be used to highlight the effect of climate change and the need to take action at an individual level.
6. Messages should focus on the local impact of climate change. Graphics showing the local potential impacts of climate change over the short and long term are likely to be more effective than more complicated maps, graphs and plans.

The report suggested that the UKCP09 projections need to be handled carefully if they are used for communicating with people who are not familiar with climate science. In many cases, climate change appeared to be, in and of itself, one of the key barriers to engagement. A question that arose from the research was: Does it matter whether people accept that climate change per se could have implications for their coastline, provided they accept that future risks exist and they engage, mitigate and adapt accordingly?

In light of this, the study stressed that it is crucial that communicators fully understand what they are trying to communicate. In many cases overemphasising the projections themselves can create a barrier to the effective communication of risk from weather and climate, although the science is crucial in underpinning the messages delivered to communities.

Difficulties & limitations

It was at times difficult to explain to some stakeholders how this research fitted in among lots of other coastal related projects and communication activities that were being undertaken. Specifically the team found it difficult to keep some consultees focused on UKCP09 and climate change projections in general.

Lessons learned

- UKCP09 climate change projections are a crucial part of the evidence base for communicating with communities, but do not need to figure extensively in *initial* messages. Experts need to keep messages simple and local with references for example to measured sea level rises along that specific piece of coast and any supporting photographs.
- Key challenges to communicating effectively on the implications of climate change include convincing people to accept that future risk is not absolute

fact, nor is it just vague opinion - it is based on a series of highly developed and informed scientific assessments.

- It was found that it can be difficult to convey messaging and engage with some stakeholders if the UKCP09 wording is used verbatim. In engaging the public, the project found it helpful to tailor the messages in UKCP09 appropriately for intended audiences.

How will the result be communicated?

The results are presented in a full project report published by Defra. The report details the research approach and methodology, trialling different approaches to stakeholder engagement and conveying messages about climate change to local communities and related stakeholders.

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- [Download the full project report](#) [3689 kb] [31/01/2011]

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